WEAKLY ETALE MORPHISMS TESTED ON AFFINE OPENS, PLUS RESTRICTION TO OPEN SUBSCHEMES

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 18934-18954 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 18956-18971, the complete author proof (14 body lines).

Editorial note. Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Fourth member of the weakly etale series (trilogy collected as L16089-L16091); the statement-side stray subscript Y is the pinned original's own and stays.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Definition of weakly etale morphism and ring map (chapter definition section plus Algebra definition), Schemes, lemma open-fibre-product; Morphisms, lemma flat-characterize (flatness local on source and target and the affine module dictionary).

Source reference. Original lemma 01U5: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 18971–18971. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01JR: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 18971–18971. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $f : X \to S$ be a morphism of schemes. The following are equivalent*

1. *The morphism f is weakly étale.*
2. *For every affine opens $U \subset X$, $V \subset S$ with $f(U) \subset V$ the ring map $\mathcal{O}_S(V) \to \mathcal{O}_X(U)$ is weakly étale.*
3. *There exists an open covering $S = \bigcup_{j \in J} V_j$ and open coverings $f^{-1}(V_j) = \bigcup_{i \in I_j} U_i$ such that each of the morphisms $U_i \to V_j$, $j \in J, i \in I_j$ is weakly étale.*
4. *There exists an affine open covering $S = \bigcup_{j \in J} V_j$ and affine open coverings $f^{-1}(V_j) = \bigcup_{i \in I_j} U_i$ such that the ring map $\mathcal{O}_S(V_j) \to \mathcal{O}_X(U_i)$ is of weakly étale, for all $j \in J, i \in I_j$.*

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Moreover, if f is weakly étale then for any open subschemes $U \subset X$, $V \subset S$ with $f(U) \subset V$ the restriction $f|_U : U \to V$ is weakly-étale.

Proof. Suppose given open subschemes $U \subset X$, $V \subset S$ with $f(U) \subset V$. Then $U \times_V U \subset X \times_Y X$ is open (Schemes, Lemma 01JR) and the diagonal $\Delta_{U/V}$ of $f|_U : U \to V$ is the restriction $\Delta_{X/Y}|_U : U \to U \times_V U$. Since flatness is a local property of morphisms of schemes (Morphisms, Lemma 01U5) the final statement of the lemma is follows as well as the equivalence of (1) and (3). If X and Y are affine, then $X \to Y$ is weakly étale if and only if $\mathcal{O}_Y(Y) \to \mathcal{O}_X(X)$ is weakly étale (use again Morphisms, Lemma 01U5). Thus (1) and (3) are also equivalent to (2) and (4). □